

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Geology & Geophysics

2. **Subject Name:** Engineering Geology (Theory) **L-T-P:** 3-0-0 **Credits:** 3

3. **Pre-requisites:** NA

4. **Syllabus and reference books:**

Syllabus:

Rock failure hypotheses and criteria, Mechanical behaviors of virtually isotropic and anisotropic rock materials, Rock physical and index properties, Shear behavior of rock discontinuities, Rock mass classifications and geological strength index, Weathering and weathering classifications of rock materials and masses, In situ stresses and their measurement, Physical properties of soils and Atterberg limits, Mechanical properties of Soils, Soil compaction, One-dimensional consolidation of soils, Site investigation and soil improvement, Alkali-aggregate reaction, Slope stability and calculation of factor of safety, Slope stabilization methods, Influence of geology on tunnelling, Excavation principles, Underground excavation instability mechanisms, Parts of a dam, Types of dams, Influence of geology on dam construction, Forces acting on a dam.

Reference Books:

- 1) "Practical Engineering Geology" by Steve Hencher
- 2) "Introduction to Rock Mechanics" by R. E. Goodman
- 3) "A Geology for Engineers" by F. G. H. Blyth and M. H. de Freitas
- 4) "Soil Mechanics and Foundations" by Muni Budhu
- 5) "Engineering Geology" by F. G. Bell
- 6) "Geology for Ground Engineering Projects" by Chris J. N. Fletcher
- 7) "Engineering in Rocks for Slopes, Foundations and Tunnels" Edited by T. Ramamurthy

5. **Lecture-wise break-up:**

Sl. No.	Topic	No. of lectures
	Basics of engineering geology	
1	Rock failure hypotheses and criteria	3
2	Physico-mechanical properties of rock materials	6
3	Rock mass characterization	3
4	Weathering classification and its implications	4
5	In-situ stresses and their determination	3
6	Physico-mechanical behaviour of soil materials	6
7	Site investigation and ground improvement	3
8	Application of engineering geology to rock and soil engineering	9
Total number of hours		37